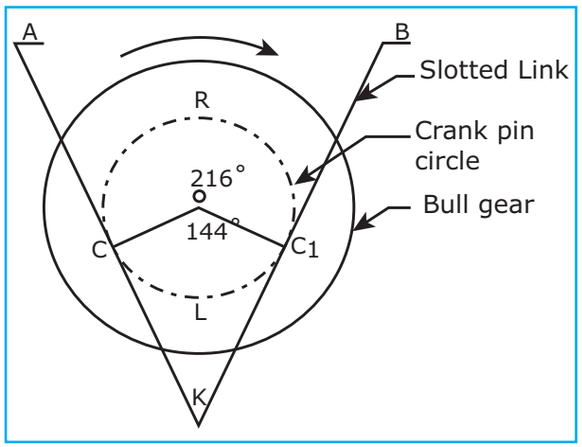
the top of the slotted link is connected to the ram, the ram reciprocates horizontally. So, bull gear rotation is converted into the reciprocating movement of the ram.

Quick return mechanism

As shown in the diagram, 'KA' indicates the starting point of the forward cutting stroke and 'KB' the end of the cutting stroke. The rotation of the crank 'OP' in clockwise direction through the angle CRC_1 refers to the forward cutting stroke. The rotation of the crank in the same direction through the angle C_1LC refers to the return stroke. As the angle C_1LC is smaller than the angle CRC_1, the time taken for the return stroke is less than that of forward stroke. So, it is evident that the speed at which the ram travels during return stroke is more.



Quick Return Mechanism

Time taken for forward cutting stroke	=	angle CRC_1
Time taken for the idle return stroke	=	angle C_1LC
	=	216^0 / 144^0
	=	3 / 2,

In some machines this ratio can be set as 7/5.

The strokelength of a ram is the distance the ram moves forward or backward. It depends upon the distance between the centre of the bull gear and the centre of the sliding block. It is adjusted according to the length of the work.

3.6 The size of a shaper

The size of a shaper is determined by the maximum length of stroke it can make. Shapers with maximum strokelength of 175 mm to 900 mm are available. Machines with maximum strokelength of 300 mm 450 mm and 600 mm are used widely.

To specify the machine further, the following points are to be provided.

1. The type of drive
   a. Individual motor
   b. Belt driven
2. The method of obtaining different speeds
   a. Gear box
   b. Step cone pulley
3. Horse power of the motor
4. Cutting to return stroke ratio
5. Number and range of speed arrangement
6. The type of the table.

3.7 Adjusting the length of stroke

The length of the stroke is calculated to be nearly 30 mm longer than the work. The position of stroke is so adjusted that the tool starts to move from a distance of 25 mm before the beginning of the cut and continues to move 5 mm after the end of the cut. For example as shown in Figure, the length of the work is 100 mm. The

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